$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\rho}_{\mathcal{S}}(t) = -\mathrm{i}\big[\hat{H}_{\mathcal{S}}, \hat{\rho}_{\mathcal{S}}(t)\big] \\ - \int_0^\infty \mathrm{d}\tau \, \Big\{ \mathcal{C}(\tau)\big[\hat{X}, \hat{X}(-\tau)\hat{\rho}_{\mathcal{S}}(t)\big] + \mathcal{C}(-\tau)\big[\hat{\rho}_{\mathcal{S}}(t)\hat{X}(-\tau), \hat{X}\big] \Big\} . \quad (5.21)$$